Lab 11: Inheritance — Genetics Problems

BIOL-1

Name: ______________________	**Date:** ______________________

Overview

Mendelian genetics allows us to predict the probability of traits in offspring. In this lab, you will apply the laws of segregation and independent assortment to solve genetics problems, analyze pedigrees, and understand inheritance patterns beyond simple dominance.

Learning Objectives

By the end of this lab, you will be able to:

1. Use Punnett squares to predict genotypic and phenotypic ratios.
2. Analyze pedigrees to determine if a trait is autosomal/sex-linked and dominant/recessive.
3. Calculate probabilities for monohybrid and dihybrid crosses.
4. Interpret blood type compatibility using the ABO system.

Activity 1: Monohybrid Cross Practice

Problem: In pea plants, Purple flowers (P) are dominant to White flowers (p).
Cross: Heterozygous Purple (Pp) × White (pp).

1. Set up the Punnett square below.
2. What is the genotypic ratio?
3. What is the phenotypic ratio?

Activity 2: Dihybrid Cross — Independent Assortment

Problem: In guinea pigs, Black hair (B) is dominant to Brown (b), and Short hair (S) is dominant to Long (s).
Cross: Heterozygous for both traits (BbSs) × Heterozygous for both traits (BbSs).

1. What are the possible gametes for a BbSs parent? (Hint: FOIL — First, Outer, Inner, Last)
2. Fill in a 16-square Punnett square (on separate paper).
3. How many offspring (out of 16) are **Black with Short hair**?

Activity 3: Pedigree Analysis

Review the family tree provided in the handout.

- **Symbols:** Squares = Male, Circles = Female. Shaded = Affected.

Questions:

1. Does the trait appear in every generation?
2. Do unaffected parents have affected children? (If yes, it's likely recessive).
3. Are significantly more males affected than females? (If yes, likely X-linked).

Based on the pedigree, is the trait Autosomal Dominant, Autosomal Recessive, or X-Linked Recessive? Explain your reasoning.

Activity 4: ABO Blood Types

Problem: A mother has Type A blood (genotype $I^A i$) and a father has Type B blood (genotype $I^B i$).

1. What are the possible blood types of their children?
2. Can they have a child with Type O blood? Show the Punnett square.

Conclusion

Genetics is a game of probability. By understanding how alleles separate and recombine, we can predict the likelihood of traits and track diseases through families.